

Alpha Without a Peer Group: Self-Referential Factor Investing in Perpetual Futures

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Abstract

Cross-sectional factor investing rests on an assumption rarely stated: that assets can meaningfully be ranked against one another. Equities supply the anchor that makes this reasonable—shared accounting, industry structure, cash flows. Cryptocurrency perpetual futures supply none of it. We ask what happens to factor construction when the comparability anchor is absent. Transporting ten factors from the published equity literature to 112 perpetual contracts over a 363-week panel, we find that ranking each contract against its own trailing history dominates ranking it against its peers in nine of eleven cases, raising mean annualised Sharpe from 1.01 to 1.45; with the carry component removed entirely the effect persists at eight of eleven. We decompose the mechanism: standardising each asset against its own history before ranking cross-sectionally recovers 41% of the gap, identifying scale heterogeneity as one channel and distributional robustness as the remainder. We further show that preserving factor independence through construction—forming one book per factor and combining returns rather than scores—adds 0.665 Sharpe ($t = 5.82$) when both constructions are tuned over an identical grid of 1,286 configurations. We introduce a factor exploiting aggressor-side information that centralised venues publish exactly and equity research must infer; it ranks third of eleven by leave-one-out contribution. Results survive walk-forward selection across seven rebalance anchors, a held-out universe test with a matched-breadth control, costs at four times the modelled level, and a deflated Sharpe correction whose trial count is read from an executed experiment log.

Keywords: factor investing, cross-sectional ranking, cryptocurrency, perpetual futures, portfolio construction, backtest overfitting

JEL: G11, G12, G14, C58

[PLACEHOLDER — COMPLETE BEFORE UPLOAD] SSRN requires a disclosure statement, displayed with the abstract, where AI tools were used in preparing the work.

1 Introduction

Every cross-sectional factor model begins with a comparison. We say an asset is cheap, or small, or heavily traded, and each term carries an implicit completion: *relative to its peers*. The comparison is defensible in equities because the market supplies scaffolding that makes it meaningful. Firms report under common accounting standards, occupy classifiable industries, and are valued against cash flows that exist. Ranking them compares like with like.

Cryptocurrency perpetual futures dissolve that scaffolding. There are no earnings, no industry classification, and no accounting identity tying price to anything. Whether one contract’s turnover is high *relative to another’s* is a question the data cannot really answer, because the quantities were never commensurate. Yet the factor literature, in turning to this market, has largely imported the cross-sectional apparatus unchanged.

This paper asks what that assumption costs, and what replaces it.

Our design is deliberately conservative. Rather than proposing new signals, we transport ten factors established in the published equity literature to 112 perpetual contracts across a 363-week panel, retaining their original definitions and sign conventions. Nothing is estimated. The only quantity we vary is the frame of comparison: in one specification each contract is ranked against its peers within the week, as the literature does; in the other, against its own trailing history—asking not whether an asset is unusual relative to others, but whether it is unusual *for itself*.

First, the comparison frame matters, and the effect is not an artefact of carry. Self-referential ranking wins in nine of eleven factors, raising mean annualised Sharpe from 1.011 to 1.451. Because we find the funding-rate component contributes 0.5–1.0 Sharpe to every book—a magnitude we report as a finding in its own right—we repeat the comparison with that component removed entirely. It survives: eight of eleven factors, with a gap of +0.265.

Second, we identify a mechanism rather than only an effect. A controlled intermediate specification, standardising each asset against its own trailing window before ranking cross-sectionally, recovers 41% of the gap. Scale heterogeneity is therefore one channel but not the dominant one. We attribute the residual to distributional robustness: a rank transform is invariant to the shape of the distribution it is applied to, whereas standardisation normalises only the first two moments and implicitly assumes approximate normality—an assumption these distributions violate severely.

Third, construction dominates selection. Standard practice blends factor scores into a composite and forms one portfolio from the induced ordering. This compresses several near-independent signals into a single ranking and discards their independence before any position is taken. Forming one book per factor and combining the resulting *returns*, holding every hyperparameter fixed, improves performance by 0.665 Sharpe ($t = 5.82$), favourable in 15 of 17 matched comparisons. Both constructions were swept over an identical grid of 1,286 configurations; comparing a tuned method against an untuned baseline is the most common way this literature manufactures

a result, and we take care not to.

Fourth, we exhibit a factor whose construction depends on information this market publishes and equity research must infer. Work on Chinese A-shares established that predictive content resides in the *shape* of the per-bar trade-size distribution rather than its mean. That measure is necessarily direction-blind, since A-share tick data does not reliably identify the initiating side. Perpetual futures publish taker-side volume on every bar, permitting the distribution to be partitioned by initiating side and its asymmetry measured. We do not overstate the novelty: US equity researchers can infer direction via the tick rule (Lee and Ready 1991), with known error rates. The distinction is measured versus estimated. The resulting factor ranks third of eleven by leave-one-out contribution.

Throughout we hold ourselves to the significance standard of Harvey et al. (2016), who argue that the proliferation of published factors implies a hurdle of $t > 3.0$ rather than the conventional $t > 2.0$. We report where we clear it and where we do not. Table 1 summarises every statistical test applied in this paper, the threshold each is judged against, and the outcome; individual tests are developed in Sections 5–7.

Finally, this paper reports an audit of its own claims. During preparation we searched the research codebase for the executable artifact behind every headline figure; four had none, including a pair of statistics that proved to be tail-index values transcribed from a document about an unrelated strategy. Section 8 reports this alongside the infrastructure now preventing recurrence. We regard the reliability of a reported backtest as an empirical property of the pipeline producing it, and therefore as something to be tested.

2 Related literature

Cross-sectional factor construction. The characteristic-sorted portfolio is the workhorse of empirical asset pricing (Fama and French 1993), and the debate over whether characteristics or covariances drive returns (Daniel and Titman 1997) presumes throughout that assets are commensurate enough to sort. We take that presumption as the object of study rather than as background.

Time-series versus cross-sectional signals. Moskowitz et al. (2012) established that an asset’s own past return predicts its future return independently of any cross-sectional ordering. Our ranking primitive is the same statistical object; the contribution is not the primitive but its systematic application across eleven factors spanning price, flow, positioning and trade-size distribution, with a decomposition of why it works.

Cryptocurrency asset pricing. Recent work constructs cross-sectional factor models for digital assets, including four-factor specifications estimated on large panels. That literature builds a *cross-sectional* model, which makes Section 5.1 a direct empirical response rather than a parallel exercise.

Backtest overfitting. Bailey and López de Prado (2014) formalise deflation of an observed Sharpe by the expected maximum across trials; Harvey et al. (2016) raise the significance hurdle; Hoffstein et al. (2018) document that rebalance-schedule choice alone generates dispersion often exceeding the effect under study. We apply all three, and in the first case address the standard weakness that the trial count is supplied by the researcher.

3 Data

Universe. 112 USDT-margined perpetual futures contracts, screened for continuous trading and finite positive prices, spanning 363 weeks. After factor warm-up the strategy is evaluable on 308 weekly rebalances and active on 282 (Table 2).

Sources. Price and volume derive from hourly bars in a public bulk archive, resampled weekly. Intraday inputs are reduced from 1-hour and 5-minute bars to one record per UTC day. Positioning data—open interest, top-trader and all-account long/short ratios—comes from a keyless exchange metrics endpoint at daily frequency. No proprietary or paid data is used.

Two venue properties are load-bearing. First, aggressor side is published: every bar carries taker-buy volume, where equity microstructure research must infer direction (Lee and Ready 1991). Second, funding is observable and material—a price-only backtest of a long/short book here overstates returns by approximately 3.9% annually. We subtract realised funding at every rebalance.

A resolution constraint discovered by failure. Trade-size distribution factors cannot be computed hourly: a day divides into roughly twelve bars per side, below the twenty-observation minimum the shape estimator requires. Measured computability is 0% at 1-hour and 100% at 5-minute. The remaining intraday factors measured *better* hourly. The pipeline therefore runs at two resolutions simultaneously—a constraint the data imposed, not an optimisation.

4 Methodology

4.1 Ranking primitives

Let $f_{i,t}$ denote a raw factor observation for asset i at time t . The cross-sectional score ranks within a date,

$$s_{i,t}^{\text{XS}} = \text{rank}(f_{i,t} \mid \{f_{j,t}\}_{j=1}^N),$$

meaningful only if the $f_{j,t}$ are commensurate. The self-referential score ranks within an asset,

$$s_{i,t}^{\text{TS}} = \text{rank}(f_{i,t} \mid \{f_{i,\tau}\}_{\tau=t-w}^{t-1}),$$

making no cross-asset comparison. Both map to $[-1, +1]$ via $((\text{rank} - 0.5)/n - 0.5) \cdot 2$, which is centred, antisymmetric and NaN-preserving. We use $w = 52$ weeks with a 26-week minimum.

The two degrade differently. As the cross-section becomes more heterogeneous, s^{XS} increasingly ranks *asset identity* rather than *signal*: a contract whose order flow is systematically larger than another’s occupies the same tail week after week irrespective of new information. s^{TS} is immune by construction.

4.2 Portfolio formation

Each factor independently forms a dollar-neutral quintile book—long the top 20%, short the bottom 20%, weights $\pm 1/n$. Scores are never blended. Book *returns* are combined by inverse trailing volatility, $w_t^{(k)} \propto 1/\hat{\sigma}_{t-1}^{(k)}$ normalised to sum to one, with the estimate lagged so no contemporaneous information enters. Where the risk model is not estimable the book is held flat rather than traded equally weighted.

4.3 Carry and turnover

Each score is penalised by the contract’s cross-sectional funding rank—the one place a cross-sectional comparison is appropriate, since funding is denominated identically across contracts. Section 5.3 shows this component is far more consequential than the term “tilt” suggests.

A uniform partial adjustment caps per-rebalance turnover. At the modelled maker fee this *costs* performance monotonically; what it purchases is robustness to cost misestimation. The deployed configuration therefore uses a cap that is not Sharpe-maximising—a deliberate divergence we state rather than leave to be discovered.

4.4 Return construction

$$r_t = \underbrace{w_t' \text{fwd}_t}_{\text{price}} - \underbrace{w_t' \text{funding}_{t+1}}_{\text{carry paid}} - \underbrace{|w_t - w_{t-1}|' c}_{\text{transaction cost}}$$

where c is a liquidity-scaled maker fee from 1 basis point for the most liquid quintile to 5 for the least, interpolated on cross-sectional dollar-volume rank. Forward returns are clipped at +100% so a single listing spike cannot manufacture a Sharpe.

4.5 Point-in-time discipline

Every factor input is leak-tested: a historical score is recomputed with all future observations deleted and required to reproduce exactly. This test exists because it previously *failed*. A ranking helper applied along the wrong array axis ranked each observation against the asset’s entire history, future included, altering historical scores for 98 of 112 contracts. Research and production code agreed perfectly throughout, because both called the same incorrect function. All results here postdate the correction and pass at 0.00e+00.

5 Empirical results

5.1 The comparison frame

Table 3 reports eleven single-factor books under both ranking frames, holding data, universe, costs and construction fixed, both as the strategy trades it and with carry removed entirely.

As traded, mean annualised Sharpe rises from 1.011 to 1.451, nine of eleven factors improving. With carry removed it rises from 0.383 to 0.648, eight of eleven improving. **The ranking effect is independent of carry:** the component amplifies the gap but does not create it. No factor changes sign under either specification, so the appropriate claim is degradation of the cross-sectional frame, not its failure. One factor reverses the sign of its ranking preference when carry is removed; at this noise level individual reversals should not be interpreted, and only the aggregate is meaningful.

5.2 Mechanism

Table 4 introduces an intermediate specification: standardise each asset against its own trailing window, then rank cross-sectionally. If scale heterogeneity drives the effect this should recover it. It recovers 41%.

Scale heterogeneity is therefore one channel but not the dominant one. We attribute the residual to distributional robustness: standardisation normalises only the first two moments, whereas a rank against an asset’s own history is invariant to the entire distributional shape. Tests separating the fat-tail component from higher moments are left to future work.

5.3 Carry is a factor, not an adjustment

Table 5 reports what removing the funding component costs each book: between +0.540 and +1.048 annualised Sharpe. Traded alone on the matched sample, carry returns 1.250 ($t = 2.91$). Carry is a first-order signal in perpetual futures. Every single-factor figure in this paper includes it, and we state so rather than presenting those numbers as the ranking effect in isolation.

5.4 Construction dominates selection

Table 6 reports the construction comparison two ways. The paired test holds every hyperparameter fixed and varies only construction across 17 matched cells: 1.657 against 0.992, a mean difference of +0.665 with $t = 5.82$, favourable in 15 of 17. The best-of-grid comparison sweeps all six variants over the same grid; the proposed construction ranks first on both maximum (2.571) and mean (1.537), the conventional last (1.971 and 0.995).

5.5 Factor contributions

Table 7 reports leave-one-out contributions on a matched 282-week sample—the full book minus each factor, versus the full book. This is the quantity of interest for a portfolio member, as distinct from standalone performance.

The trade-size asymmetry factor ranks third of eleven, and its contribution is present in *both* halves of the sample (+0.030 and +0.079), indicating structure rather than a single-period artefact. Separately, **four factors carry negative contributions**: the book is marginally better without them. We report the eleven-factor specification because it was fixed a priori; removing the underperformers post hoc would be a selection decision made with knowledge of the outcome, and we decline to make it.

6 Robustness

6.1 Walk-forward with rebalance-anchor variation

Configuration is selected on a trailing 104-week window and evaluated on the following 26 weeks, rolling forward. Because a weekly strategy also carries an unmeasured exposure to *which weekday* it rebalances (Hoffstein et al. 2018), the panel is rebuilt from hourly bars on each of seven anchors and the procedure repeated (Table 8).

Mean out-of-sample Sharpe is 2.068, all seven anchors profitable, 182 out-of-sample weeks per anchor. We report the mean rather than the maximum, and state explicitly that Monday—the schedule actually traded—is the most favourable of the seven. Six of seven anchors clear the Harvey et al. (2016) hurdle of $t > 3.0$; Friday, at $t = 2.95$, does not.

6.2 Fold geometry

Table 9 varies the train/test split from 78/13 to 156/26. All seven geometries produce profitable out-of-sample series with t from 3.70 to 5.55, every one clearing $t > 3.0$. The geometry used for the headline is mid-range rather than maximal.

6.3 Held-out universe

Time-series testing does not establish that hyperparameters generalise across *assets*. We hold out symbols: configuration is selected on 78 contracts and evaluated on 34 that had no influence on the choice, across eight random splits, with the cross-section rebuilt within each side so ranks, quintile boundaries and the cost model are computed only among symbols present.

A naive comparison is misleading here, and we flag it because we initially made it. A 34-contract book is mechanically inferior to a 78-contract one, since information ratio scales with the square

root of breadth (Grinold 1989). Table 10 therefore adds a matched-breadth control: the same selected configuration evaluated on a random 34 of the *training* contracts.

All eight splits are profitable out of universe. More informative than the ratio is the stability of the selection: all eight chose identical funding weight, turnover treatment and quintile width, and five of eight the same ranking window; fitted noise would not converge this way. We note that only one of eight held-out splits reaches $t > 3.0$, a consequence of reduced breadth, and that the held-out and control distributions overlap substantially. The defensible claim is that retention is materially above 50%, not precisely 83%.

6.4 Multiple testing with a measured trial count

The deflated Sharpe ratio (Bailey and López de Prado 2014) requires the number of trials and their dispersion. Both describe the *search* rather than the strategy and cannot be recovered from the winning backtest, so in practice they are supplied from recollection—precisely the quantity the correction exists to discipline. We instead log every evaluated configuration to an append-only registry, written by the code performing the evaluation, and read both inputs from it (Table 11).

We report both readings. The defensible choice is $N = 211$, since roughly 1,075 of the logged cells constitute the construction ablation executed *after* the strategy was specified, to measure the baseline fairly rather than to select among candidates. The conservative reading falls below 0.95 and we state it. Its instability is itself the argument against it: the value fell from 0.9507 to 0.9475 purely because an ablation continued running. A statistic that penalises conducting additional controlled experiments is not measuring selection bias.

6.5 Transaction costs

Table 12 scales the entire liquidity-adjusted cost curve. The uncapped book is superior only while realised costs remain near the assumption, deteriorating sharply beyond roughly four times it; the deployed configuration remains viable at eight times.

6.6 Research–production equivalence

The relationship between a published backtest and the system trading it is normally asserted. We test it. Research and live paths call a single shared implementation, and an automated gate asserts point-in-time integrity, determinism, dollar-neutrality, non-emptiness and identical position directions, exiting non-zero so it can block deployment (Table 13).

7 Economic significance and capacity

A Sharpe ratio is not an investment. Table 14 estimates deployable capital from realised liquidity, the book’s measured breadth (76 concurrent positions) and measured weekly turnover (0.369 of gross), under a participation constraint on the 25th-percentile contract by average daily dollar volume—the binding constraint, since the book must trade its illiquid members as well as its liquid ones.

At 2% participation the strategy supports approximately \$67 million; at 5%, approximately \$168 million. These are order-of-magnitude bounds, not precise limits: they assume participation scales linearly with capital and ignore that impact is convex in size, so the true limit is lower. They also inherit the maker-fee assumption, which Section 6.6 shows the deployed configuration survives to eight times—bounding, rather than resolving, the concern.

8 What did not survive

8.1 An audit of our own claims

Before submission we searched the research codebase for the executable artifact behind every headline figure. Four had none (Table 15). A separate claim—that one factor reverses sign between ranking frames—was tested directly and failed: zero of eleven reverse.

We report this because the corrected figures appear in this paper and the discarded ones do not, and because the episode motivates the infrastructure in Sections 6.4 and 6.6.

8.2 Negative results

We conjectured that crypto factors are structurally more orthogonal than equity factors. Measured pairwise correlation among Alpha101-style books is +0.2173 in this market against a reported 0.159 in equities: factors here are *more* correlated, not less. The orthogonality that makes per-factor construction valuable derives from data-source diversity—price, flow, positioning, trade-size distribution—not from the asset class.

One candidate factor produced a Spearman information coefficient of +0.0506 ($t = 6.10$) alongside a Pearson coefficient of +0.0172 and a decile table with no monotonic gradient. Returns here are severely fat-tailed; a factor can order the typical asset correctly while being wrong about the few observations that dominate realised profit. We report Pearson alongside Spearman throughout.

9 Conclusion

Factor investing imported into cryptocurrency perpetual futures inherits an assumption this market does not support: that its constituents are comparable. We find the assumption costly but not fatal. Cross-sectional ranking remains profitable, while ranking each contract against its own history performs materially better in nine of eleven cases—eight of eleven with carry removed—for reasons we partly decompose. We further find that how signals are combined after ranking matters more than any individual signal, and we exhibit a factor whose construction depends on information this market publishes exactly and equity research must infer.

We emphasise the structure of the evidence as much as any single result: out-of-sample in time and across assets, replicated over seven rebalance schedules and seven fold geometries, corrected for multiple testing using a trial count read from an executed log, stress-tested to eight times modelled costs, and verified to be computed identically by the deployed system. Each test was capable of overturning the result.

Limitations. Our evidence covers one venue over roughly seven years. Out-of-sample testing is conducted in time and across the cross-section, but not across venues. Capacity is estimated under a linear participation assumption that understates convex impact. One factor’s sign convention was inverted with knowledge of the data. The held-out universe result rests on eight splits. Four of the eleven factors contribute negatively on leave-one-out.

Tables

Pre-submission checklist

- Cite and differentiate the Management Science crypto factor-pricing literature; Section 5.1 is a direct empirical response.
- Insert the AI-use disclosure with the abstract (SSRN requirement).
- Data and code availability statement (Management Science requirement).
- Anonymise for Management Science: remove name, affiliation, acknowledgements; strip PDF metadata.
- Appendix: nineteen rejected constructions, enumerated.
- Appendix: 100-word plain-language abstract for the practitioner version.

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Table 1: Summary of statistical evidence. Every test applied in this paper, the threshold it is judged against, and the outcome.

test	statistic	threshold	outcome
In-sample significance			
t -statistic, full book	$t = 5.03, n = 282$	$t > 3.0$ (Harvey)	pass
Probabilistic Sharpe vs zero	PSR = 1.0000	> 0.95	pass
Multiple-testing correction			
Deflated Sharpe, strategy search	0.9885, $N = 211$	> 0.95 (Bailey)	pass
Deflated Sharpe, all logged cells	0.9475, $N = 1,286$	> 0.95	fail
Out-of-sample in time			
Walk-forward, anchor-averaged	$SR = 2.068$	> 0	pass
Walk-forward, per anchor	$t = 2.95\text{--}5.08$	$t > 3.0$	6 of 7
Walk-Forward Efficiency	103%	$> 50\text{--}60\%$	pass
Fold-geometry robustness	$t = 3.70\text{--}5.55$	$t > 3.0$	7 of 7
Out-of-sample across assets			
Held-out universe, 8 splits	8 of 8 profitable	> 0	pass
Retention at matched breadth	83%	$> 50\%$	pass
Held-out per-split t	1 of 8 reach $t > 3.0$	$t > 3.0$	1 of 8
Rebalance-schedule luck (Hoffstein)			
Timing luck, 7 anchors in-sample	$t = 3.67\text{--}5.03$	$t > 3.0$	7 of 7
Anchors profitable	7 of 7	—	pass
Construction effect			
Paired difference, 17 cells	$+0.665, t = 5.82$	$t > 3.0$	pass
Cells favouring construction	15 of 17	$> 50\%$	pass
Individual components			
Trade-size asymmetry, standalone	$t = 3.35$	$t > 3.0$	pass
Trade-size asymmetry, subperiods	$+0.030 / +0.079$	both > 0	pass
Carry, standalone	$t = 2.91$	$t > 3.0$	fail
Cost and capacity			
Cost robustness, deployed book	viable to $8\times$	$\geq 1\times$	pass
Capacity at 2% participation	\$67m	—	—
Implementation integrity			
Point-in-time leak, all factors	0.00e+00	$= 0$	pass
Dollar-neutrality	9.02e−17	≈ 0	pass
Research = live directions	0 of 4,480	$= 0$	pass
Determinism	exact	exact	pass

Table 2: Sample and universe.

Universe	112 USDT-margined perpetual futures
Panel	363 weeks
Rebalances evaluable / active	308 / 282
Rebalance frequency	weekly
Factors	11 (10 transported, 1 introduced)
Cost model	liquidity-scaled maker fee, 1–5 bp
Funding	subtracted at every rebalance

Table 3: Ranking frame. Eleven single-factor books, annualised Sharpe, net of funding and costs. “Carry removed” sets the funding weight to zero.

specification	cross-sectional	self-referential	gap	self wins
as traded (with carry)	1.011	1.451	+0.440	9 of 11
carry removed	0.383	0.648	+0.265	8 of 11
sign reversals	0 of 11 under either specification			

Table 4: Mechanism decomposition.

specification	mean annualised Sharpe	gap recovered
cross-sectional, raw	1.011	—
cross-sectional, per-asset standardised	1.192	41%
self-referential	1.451	100%

Table 5: Carry is a factor. Single-factor books with and without the funding component.

factor	with carry	without carry	carry contributes
TSKD	1.442	0.394	+1.048
TKU	1.494	0.674	+0.820
OFI	1.241	0.480	+0.762
AVOL	1.222	0.663	+0.559
Quad	1.896	1.356	+0.540
<i>carry traded alone, matched sample: 1.250 ($t = 2.91$)</i>			

Table 6: Construction. Both constructions swept over an identical grid of 1,286 configurations.

Panel A: paired comparison (hyperparameters fixed)		
matched cells	17	
one book per factor, self-referential	1.657	
blended scores, cross-sectional, equal weight	0.992	
mean difference	+0.665**	
t-statistic of difference	5.82	
cells favouring proposed construction	15 of 17	
Panel B: best-of-grid, all six variants		
construction	best	mean
self / books / risk parity	2.571	1.537
self / blend / equal	2.468	1.288
self / books / equal	2.296	1.510
cross / books / equal	2.200	1.152
cross / books / risk parity	2.113	1.129
cross / blend / equal	1.971	0.995

Table 7: Leave-one-out factor contributions. Matched 282-week sample; full eleven-factor book returns 2.161 ($t = 5.03$).

factor	book without it	contribution
Q	2.033	+0.128
TKU	2.084	+0.078
TSKD	2.115	+0.046
WRspread	2.123	+0.038
AVOL	2.135	+0.026
OFI	2.144	+0.018
TopChg	2.150	+0.011
CPVv	2.170	−0.008
RSJ	2.179	−0.017
Quad	2.180	−0.019
CPVm	2.189	−0.028

Table 8: Walk-forward out-of-sample Sharpe by rebalance anchor. 182 out-of-sample weeks per anchor. Harvey et al. (2016) hurdle is $t > 3.0$.

anchor	Mon	Sat	Sun	Tue	Wed	Thu	Fri	mean
OOS Sharpe	2.713	2.496	2.423	1.855	1.753	1.656	1.579	2.068
<i>t</i>	5.08	4.67	4.53	3.47	3.28	3.10	2.95	
clears $t > 3$	yes	yes	yes	yes	yes	yes	no	6 of 7

Table 9: Fold geometry. Monday anchor; only the train/test split varies.

train	test	folds	n_{oos}	OOS Sharpe	t
78	13	16	208	2.257	4.51
78	26	8	208	1.850	3.70
104	13	14	182	2.968	5.55
104	26	7	182	2.713	5.08
104	52	3	156	2.859	4.95
130	26	6	156	3.007	5.21
156	26	5	130	2.985	4.72

Table 10: Held-out universe, eight random splits. The matched-breadth control evaluates the same selected configuration on a random 34 of the *training* contracts, isolating overfitting from the mechanical breadth effect.

	train (78)	held-out (34)	control: seen (34)
mean annualised Sharpe	2.204	1.127	1.366
median	2.169	1.194	1.407
standard deviation	0.212	0.286	0.281
splits profitable	—	8 of 8	—
held-out / train	51%	<i>conflates two effects; not interpretable</i>	
control / train	62%	<i>breadth, mechanical</i>	
held-out / control	83%	overfitting, at matched breadth	

Table 11: Deflated Sharpe ratio. Trial count and dispersion read from an executed experiment registry rather than supplied by the researcher.

trial set	N	sd(SR)	SR^*	DSR
strategy-search configurations	211	0.0768	0.2139	0.9885
all logged configurations	1,286	0.0766	0.2547	0.9475
<i>selected: $T = 269$, weekly SR 0.3565, skew +0.145, kurtosis 4.391</i>				
<i>unconditional: $\text{PSR}(SR^*=0) = 1.0000$</i>				

Table 12: Transaction-cost robustness. The entire liquidity-scaled cost curve is multiplied by the stated factor.

cost multiple	$1\times$	$2\times$	$4\times$	$8\times$	$16\times$	$32\times$
uncapped	2.364	2.247	2.014	1.547	0.618	−1.222
deployed (capped)	2.161	2.117	2.028	1.851	1.496	0.789

Table 13: Research–production equivalence gate. Exits non-zero on any failure.

assertion	measured	status
point-in-time leak, all factors	0.00e+00	pass
determinism	exact	pass
dollar-neutrality (max net)	9.02e−17	pass
gross exposure bound	$1.462 \leq 2.0$	pass
non-emptiness	308 built, 282 active	pass
research = live directions	0 / 4,480	pass

Table 14: Capacity. Binding constraint is the 25th-percentile contract by average daily dollar volume (\$3.26m); breadth 76 positions, weekly turnover 0.369.

participation rate	implied capacity
1%	\$33.6m
2%	\$67.2m
5%	\$167.9m
10%	\$335.8m
<i>median contract ADV: \$6.37m; 10th percentile: \$1.93m</i>	

Table 15: Claims that did not survive audit. All are excluded from this paper.

claim as previously recorded	what the audit found
walk-forward Sharpe “2.68–2.95”	tail-index statistics from a document concerning an unrelated strategy; not Sharpe ratios
timing-luck dispersion “ ± 0.24 ”	no script, output or artifact
deflated Sharpe “0.9966 / 0.9816”	no artifact
conventional baseline “1.03”	no artifact; approximately the conventional construction’s <i>mean</i> (0.992) compared against the proposed construction’s <i>maximum</i>
one factor reverses sign between frames	tested directly; zero of eleven reverse